

## Water-based battery that will not catch fire



Water-based zinc batteries (Credit: University of Maryland)

A team of researchers from University of Maryland's (UMD's) A. James Clark School of Engineering have created a water-based zinc battery that is simultaneously powerful, rechargeable and intrinsically safe. The new aqueous zinc battery could eventually be used in consumer electronics, as well as in extreme conditions to improve the performance of safety-critical equipment and vehicles used in aerospace, military and deep-ocean environments.

Together with colleagues at U.S. Army Research Laboratory (ARL) and National Institute of Standards and Technology, UMD engineers used metallic zinc—an element used in 1799 in the very first battery—to crank up the energy of their own 2015 advance in battery technology. The previous UMD advance used a novel and safe water-in-salt electrolyte to replace the flammable organic electrolyte used in conventional lithium-ion batteries.

According to Fei Wang, a jointly appointed post-doctoral associate at UMD's Clark School and ARL, and first author of the paper, "Water-based batteries could be crucial to preventing fires in electronics, but their energy storage and capacity have been limited, until now. For the first time, we have a battery that could compete with the lithium-ion batteries in energy density, but without the risk of explosion or fire."

The new highly-concentrated aqueous zinc battery also overcomes the disadvantages of conventional zinc batteries, such as the capacity to endure only limited recharging cycles, damaging growth of tree-like crystal structures (known as dendrites) during usage and recharging, and sustained water consumption that results in the need to regularly replenish the batteries' electrolyte with water.

## Solar-powered tech to purify water

Researchers are exploiting the age-old technique of using solar energy from the sun to evaporate and sanitise water. By draping black, carbon-dipped paper in a triangular shape and using it to both absorb and vaporise water, scientists from University at Buffalo have developed a method for using sunlight to generate clean water with near-perfect efficiency.

According to lead researcher Qiaoqiang Gan, PhD, associate professor of electrical engineering at University at Buffalo's School of Engineering and Applied Sciences, "Our technique can produce drinking water at a faster pace than is theoretically calculated under natural sunlight." The team increased the efficiency of its evaporation system by cooling it down.

Gan explains, "Usually, when solar energy is used to evaporate water, some of the energy is wasted as heat and lost to the surrounding environment. This makes the process less than 100 per cent efficient. Our system has a way of drawing heat in from the surrounding environment, allowing us to achieve near-perfect efficiency."

This low-cost technology could provide drinking water in



A central component of the new technology is a sheet of carbon-dipped paper folded into an upside-down V. The paper's sloped geometry keeps it cool by weakening the intensity of light illuminating it (Credit: University of Buffalo)

regions where resources are scarce, or where natural disasters have struck.